DSAA3073: Theories in Data Science

Week 6 Validation Sheet · Honest Model Selection and Evaluation

Coverage: Week 6 **Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. A team fits 20 declared candidates on training data, chooses the one with the smallest validation loss, refits it on training plus validation data, and wants a final performance estimate. Which statement is sound?

(a) The minimum validation loss is final evidence because validation data were not used for fitting.
(b) The training loss after refitting is final evidence because it uses more data.
(c) A sealed test set may evaluate the completed selection-and-refitting procedure.
(d) Averaging the training and validation minima removes selection optimism.

Solution. (c). The validation outcomes influence which candidate is selected, so the smallest validation value is selection evidence rather than untouched final evidence. Refitting on more observations does not make the training loss fresh. A sealed test set that has influenced no candidate, hyperparameter, preprocessing, threshold, or reporting decision may evaluate the completed procedure. Averaging two reused minima does not restore independence.

Marking. *5 marks for (c) and the no-feedback-to-the-sealed-test reason; 0 otherwise.*

A2 [5 marks]. In LOOCV on a dataset of size n, what does each held-out loss directly evaluate?

(a) A predictor trained on all n observations.
(b) The declared algorithm trained on $n-1$ observations.
(c) A predictor trained on one observation.
(d) A deterministic population quantity with no sampling variability.

Solution. (b). Every leave-one-out prediction is produced by a fit that excludes the scored observation and therefore uses $n-1$ observations. The folds are deterministic conditional on the observed dataset, but the score still changes across newly sampled datasets. Transferring the score to a final n-observation fit needs a stability or smooth-learning-curve qualification.

Marking. *5 marks for (b), including the $n-1$ training target; 0 otherwise.*

A3 [5 marks]. Which setting licenses the exact residual correction $e_{-i,i} = \frac{e_i}{1-h_{\lambda,ii}}$ taught in Week 6?

(a) Any algorithm whose full-data fitted values can be written as Hy.
(b) Ordinary k-NN, even when deletion changes the neighbour set.
(c) Fixed-design ordinary or ridge least squares with compatible deletion fits, unchanged features and penalty, and $1-h_{\lambda,ii} \neq 0$.
(d) Any fitted rule after its tuning parameter has been selected from the responses.

Solution. (c). The identity uses the compatible rank-one change in the ordinary or ridge least-squares normal equations. It also keeps the feature construction and penalty fixed and requires a nonzero denominator. A full-data equation $\hat{y} = Hy$ alone does not describe what the algorithm does after a case is deleted, and ordinary k -NN can change its neighbours and weights.

Marking. *5 marks for (c) and the compatible-deletion condition; 0 otherwise.*

A4 [5 marks]. Which statement correctly distinguishes the Week 6 criteria?

(a) AIC is a distribution-free finite-sample PAC bound.
(b) BIC proves that its selected model is true on any finite candidate list.
(c) AIC targets expected fresh conditional KL or log-score discrepancy under regular large-sample likelihood conditions.
(d) AIC, BIC, CV, and SRM differ only by interchangeable complexity constants.

Solution. (c). Standard AIC corrects maximized conditional log likelihood toward an expected fresh conditional KL or negative-log-score target under regular finite-dimensional maximum-likelihood asymptotics. It is not a PAC bound. BIC's model-identification reading additionally needs a regular asymptotic regime with a represented true model and suitable prior and identifiability conditions. The four methods do not share one interchangeable guarantee.

Marking. *5 marks for (c), including the predictive target and regular-asymptotic scope; 0 otherwise.*

Part B: Definition in use

B1 [20 marks]. A public-health team has 800 i.i.d. observations and a small, predeclared collection of regular Poisson regression models fitted by maximum likelihood. Every free likelihood parameter is counted. Its scientific target is fresh conditional log-score performance. The team can seal 160 observations before any model comparison, and it wants both a selected model and a numerical final performance report.

Recommend a selection-and-evaluation workflow. Justify the selection rule from the target and candidate family, trace which data may influence each decision, and explain why neither a BIC identification claim nor an SRM PAC guarantee is obtained automatically.

Solution. The target is predictive conditional log score and the candidate family is a small collection of regular finite-dimensional likelihood models fitted by maximum likelihood. Under the stated large-sample regularity and correct parameter counting, AIC is therefore a natural **selection** rule. The team should first seal the 160 observations. It computes

$$\mathrm{AIC}_j = -2\ell_{j(\hat{\theta}_j)} + 2q_j$$

using only the remaining 640 observations and chooses the lower-AIC model. Any diagnostics, transformations, or revisions triggered by those data also belong inside this selection portion.

Once the complete procedure is fixed, it is applied to the sealed 160 observations exactly once. Their mean negative conditional log score is a final performance estimate because none of those outcomes influenced the candidate list, fitted parameters, selection, or reporting rule. The AIC value itself is not substituted for that untouched report: it is a large-sample corrected training-likelihood criterion used for comparison.

BIC answers a different question. Its $q_j \log n$ correction approximates a marginal-likelihood comparison, and consistent identification would require a declared regular asymptotic regime including a true generating model in the list and suitable prior and identifiability conditions. The prompt supplies a predictive goal, not that identification goal. SRM is different again: a PAC conclusion requires a declared loss, class sequence, confidence allocation, and a proved uniform complexity penalty. A regular Poisson likelihood and an AIC computation do not create those objects. Thus the legal flow is selection on the 640-observation portion, followed by one-way evaluation on the sealed 160-observation portion, with no arrow from final outcomes back to model choice.

Marking. *4 marks for matching the predictive conditional-log-score target and regular MLE family to AIC; 4 for defining the lower-is-better AIC object with correct parameter counting; 5 for a one-way 640/160 selection-and-sealed-evaluation flow that includes all adaptive decisions; 3 for identifying what the sealed score reports; 2 for rejecting an automatic BIC identification conclusion and stating its true-model-in-list/regular-regime conditions; 2 for rejecting an automatic SRM PAC guarantee and naming its missing uniform-bound objects.*

Part C: Basic skills

C1 [7 marks]. Three cross-validation folds have sizes $2, 3, 5$ and mean held-out squared losses $1.0, 3.0, 2.0$, respectively. Compute the observation-weighted CV score. Also compute the unweighted average of the three fold means and state why it is not the required score.

Solution. The CV definition averages held-out losses over observations. The total held-out loss represented by the three rows is

$$2(1.0) + 3(3.0) + 5(2.0) = 21.$$

There are $2 + 3 + 5 = 10$ observations, so the observation-weighted score is

$$\mathrm{CV} = \frac{21}{10} = 2.1.$$

The unweighted fold-mean average is

$$\frac{1.0 + 3.0 + 2.0}{3} = 2.0.$$

It is not the required score because it gives the two-observation fold the same weight as the five-observation fold rather than giving each observation one equal contribution.

Marking. *3 marks for the weighted total 21; 2 for* 2.1*; 1 for the unweighted value* 2.0*; 1 for the unequal-observation-weight explanation.*

C2 [8 marks]. A regular Gaussian regression fitted by maximum likelihood has three regression coefficients and an estimated noise variance. At its MLE, $\ell\left(\hat{\theta}\right) = -50$, with $n = 100$ and $\log 100 \approx 4.605$. Compute its standard AIC and BIC under the lower-is-better convention. Then state in one sentence what neither number is: a PAC risk certificate or an untouched final-performance estimate.

Solution. Standard AIC and BIC count every free fitted likelihood quantity. The three regression coefficients plus the estimated variance give $q = 4$, and $-2\ell\left(\hat{\theta}\right) = 100$. Therefore

$$\mathrm{AIC} = 100 + 2(4) = 108,$$

while

$$\mathrm{BIC} = 100 + 4\log(100) \approx 100 + 4(4.605) = 118.42.$$

These are likelihood-based comparison criteria; the arithmetic does not turn either value into a PAC risk certificate or a sealed test estimate.

Marking. *2 marks for counting* $q = 4$*; 1 for* $-2\ell = 100$*; 2 for AIC* 108*; 2 for BIC* 118.42*; 1 for the criterion-not-final-guarantee interpretation.*

Part D: Construction and design

D1 [17 marks]. Consider ordinary one-nearest-neighbour regression on

$$(x_1, x_2, x_3) = (0, 1, 3), \quad (y_1, y_2, y_3) = (0, 1, 4).$$

The full-data in-sample convention includes the query point itself, so the fitted-response map is $\hat{y} = I_3 y$.

Use this dataset to construct a concrete demonstration that the least-squares PRESS residual formula is not licensed merely by a full-data linear map. Compute the actual leave-one-out predictions and score, then identify exactly what fails in the attempted PRESS calculation.

Solution. The construction exploits the difference between a full-data response map and the algorithm's behavior after deletion. With self-neighbours included, the full-data fitted values are exactly y, so $H = I_3$, every residual is $e_i = 0$, and every diagonal value is $h_{ii} = 1$. The formal PRESS expression would ask for $\frac{e_i}{1-h_{ii}} = \frac{0}{0}$, which is undefined.

The genuine leave-one-out fits are nevertheless well defined. Deleting observation 1 leaves $x_2 = 1$ as the nearest neighbour of $x_1 = 0$, so $\hat{y}_{-1,1} = y_2 = 1$. Deleting observation 2 leaves $x_1 = 0$ closer than $x_3 = 3$, so $\hat{y}_{-2,2} = y_1 = 0$. Deleting observation 3 leaves $x_2 = 1$ as nearest, so $\hat{y}_{-3,3} = y_2 = 1$. Thus the deleted residuals are

$$(-1, 1, 3)$$

and the LOOCV squared-error score is

$$\frac{(-1)^2 + 1^2 + 3^2}{3} = \frac{11}{3}.$$

The failure is structural, not just division by zero. Deleting a case changes which training points are available as neighbours and can change both the neighbour graph and its weights. The ordinary/ridge least-squares rank-one normal-equation identity that derives the PRESS correction is absent. Hence $\hat{y} = Hy$ on the full dataset is insufficient to reconstruct the deleted fits, and the actual refits, rather than the PRESS shortcut, define this LOOCV score.

Marking. *4 marks for identifying $H = I$, $e = 0$, and $h_{ii} = 1$; 3 for diagnosing the undefined $\frac{0}{0}$ PRESS expressions; 5 for all three actual deleted predictions or residuals; 2 for* LOOCV $= \frac{11}{3}$*; 3 for explaining that deletion changes neighbours and no compatible least-squares rank-one identity exists.*

D2 [18 marks]. A team adaptively compares 30 tree-based classification pipelines, including feature transformations, depths, and probability thresholds. There is no declared regular likelihood parameter count. It proposes four outer folds and five-fold CV inside every outer-training portion.

Design one complete legal inner/outer pipeline and state what the final outer average estimates. Your design must place every adaptive choice, refit, and scored observation in the correct data role, and must rule out feedback from outer outcomes.

Solution. Fix the four outer folds before examining their outcomes. For outer round r, hold outer fold E_r completely aside. Only the remaining observations may enter the adaptive inner procedure. Partition that outer-training portion into five inner folds. For every one of the 30 pipelines, each inner held-out observation must be predicted by a fit that excludes its inner fold. Average the inner losses, use those inner results to choose the feature transformation, depth, threshold, and any other data-dependent setting, and then refit the selected complete pipeline on the entire outer-training portion.

Apply that refitted pipeline once to E_r and record its classification loss. The labels or losses in E_r must not change the candidate list, inner-fold design, selected settings, stopping rule, or reporting choice. Repeat this process for all four outer folds and average the losses over outer observations (weighting by fold size if the outer folds are unequal).

This outer average evaluates the **complete inner selection algorithm** trained on an outer-training portion, not the minimum inner-CV value and not literally the final all-data fit. Transfer to the all-data refit requires the usual training-size/stability qualification. After assessment, the team may rerun the same fixed inner-selection procedure and fit the resulting pipeline on all available deployment data, but it may not revisit outer outcomes to improve the procedure while retaining the old outer average as untouched evidence. Nested CV is suitable here because it can evaluate the full adaptive pipeline without inventing a standard AIC/BIC parameter count for the trees.

Marking. *4 marks for holding each outer fold entirely outside adaptation; 5 for running all feature, model, and threshold choices inside five-fold CV on the outer-training portion; 3 for refitting the chosen complete pipeline before outer scoring; 3 for an observation-weighted outer report with no feedback; 2 for identifying the report as performance of the complete outer-training-sized selection procedure; 1 for the training-size qualification or lawful final all-data refit.*

Part E: Proof and derivation

E1 [25 marks]. A proposed regression procedure receives two candidate feature columns

$$z^{(1)} = (0, 0, 1)^T, \quad z^{(2)} = (0, 1, 1)^T$$

and a response vector $y(a) = (-a, -a, a)^T$, where $a > 0$. It first selects the column maximizing $\left|z^T y\right|$, with ties resolved in favour of $z^{(1)}$, and then fits one-dimensional ridge with $\lambda = 1$ on the selected column. An analyst takes the matrix and leverages from the full-data selected ridge fit and applies the PRESS correction to every case, claiming exact LOOCV for the entire procedure.

Give one coherent mathematical audit valid for every $a > 0$. Develop a held-out-prediction criterion, with the necessary quantifiers, for when frozen-full-selection PRESS and LOOCV obtained by rerunning the entire procedure agree; distinguish a useful sufficient selector condition from what is actually necessary. Use the displayed family to decide the analyst's claim, and then design an honest inner/outer evaluation of this adaptive procedure with a justification that no outer response can influence the prediction on which its own outer loss is computed.

Solution. There are two different deletion worlds. In the **frozen-selection world**, first select a feature on the full dataset and keep that feature fixed while deleting each case from the final ridge fit. Conditional on this fixed feature, the usual ridge PRESS identity is valid when its denominators are nonzero. In the **rerun world**, deleting case i also deletes y_i from the feature-selection step; the selector and the subsequent ridge fit are both rerun. LOOCV for the declared composite procedure belongs to the rerun world.

Write s for the full-data selected feature and s_{-i} for the feature selected after deleting case i. Let $\hat{y}_{i,-i}^{(s)}$ be the prediction at case i from ridge trained without case i while keeping feature s, and define $\hat{y}_{i,-i}^{(s_{-i})}$ analogously for the rerun selector. Assuming the fixed-feature ridge PRESS identity is licensed, frozen-selection PRESS equals rerun-procedure LOOCV case by case if and only if

$$\forall i \in \{1, ..., n\}, \quad \hat{y}_{i,-i}^{(s)} = \hat{y}_{i,-i}^{(s_{-i})}.$$

This is the necessary and sufficient criterion at the scored-prediction level. Selector stability, $s_{-i} = s$ for every i, is a convenient sufficient condition. It is not logically necessary: two different selected features could coincidentally produce the same prediction at a particular held-out point. The prediction equality, rather than equality of feature names, is the exact requirement.

We now apply this criterion to the whole family. On the full data,

$$\left|\left(z^{(1)}\right)^T y(a)\right| = a, \quad \left|\left(z^{(2)}\right)^T y(a)\right| = 0,$$

so the procedure selects $z^{(1)}$. For a one-column design z, ridge with $\lambda = 1$ has

$$H_z = z\left(z^T z + 1\right)^{-1} z^T.$$

Hence for $z^{(1)}$, $z^T z = 1$ and

$$H_{z^{(1)}} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Its fitted vector is $\left(0, 0, \frac{a}{2}\right)^T$, its third residual is $e_3 = \frac{a}{2}$, and $h_3 = \frac{1}{2}$. The analyst's fixed-full-design correction for case 3 would be

$$\frac{e_3}{1 - h_3} = \frac{\frac{a}{2}}{1 - \frac{1}{2}} = a.$$

Now delete observation 3 and rerun the **declared procedure**, not merely its final ridge equation. On observations 1 and 2, the two selection scores are

$$|(0, 0) \cdot (-a, -a)| = 0 \quad \text{and} \quad |(0, 1) \cdot (-a, -a)| = a.$$

The deletion fit therefore selects $z^{(2)}$, not $z^{(1)}$. Its training feature values are $(0, 1)$, so

$$\hat{\beta}_{-3} = \frac{0(-a) + 1(-a)}{0^2 + 1^2 + 1} = -\frac{a}{2}.$$

At the held-out third case, $z_3^{(2)} = 1$, giving prediction $-\frac{a}{2}$ and actual deleted residual

$$a - \left(-\frac{a}{2}\right) = 3\frac{a}{2}.$$

The actual residual $3\frac{a}{2}$ differs from the frozen-selection PRESS residual a for every $a > 0$.

This family exposes the quantifier error. For each fixed feature j, the terminal ridge subroutine has a response-fixed matrix H_j. But the composite rule is the piecewise map

$$y \mapsto H_{s(y)}y,$$

whose selected matrix can change with the response vector and with deletion. A conditional statement of linearity after fixing s does not provide one matrix that governs all response vectors and all deletion datasets. The displayed mismatch violates the prediction-level criterion, so the full-selected PRESS calculation is not LOOCV for the composite procedure.

For honest evaluation, fix outer folds before examining their outcomes. In outer round r, remove outer fold E_r. Every adaptive choice, including feature selection, any comparison of preprocessing rules or penalties, and all inner validation, uses only the outer-training portion. Rerun the selected complete procedure on that outer-training portion and form predictions for E_r before reading its responses for scoring. For an outer observation i, its prediction is therefore a function only of the other outer-training data and its covariates; y_i is absent from feature selection, fitting, and tuning. Only after the prediction is fixed is y_i used to compute its loss. This proves the required no-influence property. Average losses over outer observations, weighting unequal folds by their sizes, and do not feed outer outcomes back into the procedure. The result evaluates the complete adaptive procedure at the outer-training size, with the usual qualification before transfer to an all-data refit.

Marking. *4 marks for constructing the frozen-selection and rerun-procedure deletion worlds and locating fixed-feature PRESS in the correct world; 5 for a quantified necessary-and-sufficient equality of held-out predictions, together with the distinction between sufficient selector stability and actual necessity; 5 for using the full family to show selection of $z^{\{(1)\}}$ on full data, selection of $z^{\{(2)\}}$ after deleting case 3, and residuals a versus $3\frac{a}{2}$; 5 for diagnosing the conditional-subroutine versus composite-map quantifier error and applying the prediction criterion to reject the claim for every $a > 0$; 6 for an inner/outer workflow plus a valid no-influence argument for each outer observation and a correctly weighted final report. Unsupported labels such as "leakage" or "use nested CV" earn no marks without the associated representation, quantifiers, or no-influence reasoning. Preserve follow-through credit after one arithmetic slip when the structural chain is correct.*

Part F: Synthesis and limitations

F1 [15 marks]. A colleague proposes the universal identity

$$\text{generalization score} = \text{training fit} + \text{complexity penalty}$$

and says CV, AIC, BIC, and SRM differ only in the penalty. Reject or defend the proposal by comparing, for all four methods, the object actually scored, the conditions that license it, and the conclusion it can support.

Solution. The proposal must be rejected as a literal universal identity. The four methods may all react to overfitting, but their scored objects and guarantees are different.

- **Cross-validation** scores the declared loss on observations excluded from the corresponding fold fits. Its validity depends on the exclusion rule and the sampling/design assumptions; if it evaluates adaptive selection, the whole selection must run inside each training portion. It estimates performance of the fold-trained or completed selection procedure, with a stated training-size transfer qualification. It is not algebraically a full-data training loss plus a class penalty.
- **AIC** scores $-2\ell\left(\hat{\theta}\right) + 2q$ for regular finite-dimensional likelihood candidates fitted by maximum likelihood, with consistent parameter counting and large-sample conditions. It compares expected fresh conditional KL or negative-log-score discrepancy up to common constants. It is not a distribution-free PAC bound or untouched test score.
- **BIC** scores $-2\ell\left(\hat{\theta}\right) + q\log n$ under a regular marginal-likelihood approximation with suitable prior and identifiability behavior. Its leading conclusion is a marginal-likelihood ranking. Interpreting it as consistent model identification additionally needs a represented true model in the declared list and the appropriate asymptotic regime; one finite selection does not prove truth.

- **SRM** scores empirical risk plus a justified class-dependent term over a declared class sequence, loss, confidence allocation, and uniform complexity control. Its distinctive output is a stated high-probability population-risk upper bound.

Therefore even the apparent penalties $2q$, $q \log n$, and an SRM class term correct different objects under different assumptions, while CV relies on data exclusion rather than such an additive correction. A method must be chosen from the scientific target, candidate family, data flow, and theorem conditions, not by substituting one "complexity penalty" into a universal template.

Marking. *4 marks for the CV scored object, exclusion conditions, and qualified procedure-performance conclusion; 3 for AIC's regular MLE object and predictive conditional-KL/log-score target; 3 for BIC's marginal-likelihood object and conditional true-model-in-list identification reading; 3 for SRM's empirical-risk-plus-justified-class-bound object and high-probability conclusion; 2 for explicitly rejecting the literal universal identity by comparing the mechanisms rather than only naming formulas.*